EC442 Macroeconomics for MRes students | Instructor: Matthias Doepke
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Problem Set 1

Dynamic Programming: An Analytical Case

Question 1: Consider the standard growth model with a logarithmic utility function and full depreciation. The social planning problem is:

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \log(c_t) \right\}$$

subject to:

$$c_t + k_{t+1} = k_t^{\alpha},$$

with $k_0 > 0$ given.

1. Formulate a Bellman equation for the social planning problem.
2. It turns out that the Bellman equation can be solved analytically. Conjecture that the value function takes the form:

   $$v(k) = a_0 + a_1 \log(k).$$

   Solve for the coefficient a_1 and for the optimal policy function $k' = g(k)$, and find the coefficient a_0. Hint: after plugging in the conjectured functional form, you can take first-order and envelope conditions and solve the resulting system of equations.

The Standard Model

Question 2: Consider the standard optimal growth model represented by the planning problem:

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}.$$

subject to:

$$\begin{aligned} c_t + k_{t+1} &\leq (1-\delta)k_t + A_t(k_t)^{\alpha}(n_t)^{1-\alpha}, \\ A_{t+1} &= \gamma A_t, \\ n_t &= 1, \end{aligned}$$

with $\gamma > 1$, $\sigma > 0$, $\sigma \neq 1$, $\alpha, \beta, \delta \in (0,1)$, and A_0, k_0 given.

1. Consider a market structure in which a representative firm buys labor from households at wage w_t and rents capital at interest rate r_t. Define a competitive equilibrium for this economy.

2. The equilibrium turns out to be characterized by convergence to a balanced growth path. To characterize this balanced growth path, it turns out to be useful to rewrite the model in terms of variables that are detrended by productivity growth. For example, the detrended consumption $\tilde{c}_t$ is given by:

$$\tilde{c}_t = \frac{c_t}{\tilde{A}_t},$$

where $\tilde{A}_t = A_t^{1/(1-\alpha)}$. Formulate a Bellman equation for the social planning problem for this economy in terms of the detrended variables.

3. Use first-order and envelope conditions to characterize the balanced growth path (i.e., the path in which output, consumption, and capital all grow at a common rate).

4. Assuming that interest rates and wages are given by the marginal products of capital and labor, demonstrate that in the balanced growth path the economy satisfies the following observations on long-run growth:

   (a) The income shares of capital and labor are constant.
   (b) The return on capital is constant.
   (c) The capital-output ratio is constant.
   (d) The wage increases at a constant rate.

5. Assume the economy is displaced from the balanced growth path because of a partial destruction of the capital stock (say, because of an earthquake). Which of the growth facts will continue to hold on the transition path back to the balanced growth path?

The AK Model

Question 3: Consider a growth model characterized by the planning problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \right\}.$$

subject to:

$$c_t + k_{t+1} \leq (1-\delta)k_t + Ak_t.$$

Assume $\sigma \neq 1$.

1. Make a direct argument showing that the value function $v^\star(k)$ for the sequential optimization problem should be of the form:

$$v^\star(k) = a\frac{k^{1-\sigma}}{1-\sigma}.$$

2. Use the Bellman equation corresponding to the planning problem to compute a.

3. Compute the equilibrium growth rate of the economy.